MAT 5210 Midterm (Sample) — Solutions

Problem 1

(a) Let $L : K$ be an algebraic extension.

- **Separable:** for every $\alpha \in L$ the minimal polynomial $m_\alpha(x) \in K[x]$ is separable, i.e. has no repeated roots (equivalently $\gcd(m_\alpha, m'_\alpha) = 1$). In characteristic 0 this is automatic.
- **Normal:** for every $\alpha \in L$ the minimal polynomial $m_\alpha(x) \in K[x]$ splits into linear factors in $L[x]$. (For $[L : K] < \infty$: $L : K$ is normal iff L is the splitting field of some polynomial in $K[x]$.)
- **Galois:** $L^{\mathrm{Aut}_K(L)} = K$, i.e. the fixed field of the group of all K-automorphisms of L is exactly K. For $[L : K] < \infty$ this is equivalent to: $L : K$ is normal and separable; equivalently L is the splitting field of a separable polynomial in $K[x]$; equivalently $|\mathrm{Aut}_K(L)| = [L : K]$.

(b) Suppose $L : K$ is Galois. The **Galois group** is

$$\mathrm{Gal}(L/K) := \mathrm{Aut}_K(L) = \{\varphi : L \to L \ \text{ field automorphism} : \ \varphi|_K = \mathrm{id}_K\}$$

with composition as the group operation.

(c) **Fundamental Theorem of Galois Theory.** Let $L : K$ be a finite Galois extension with $G = \mathrm{Gal}(L/K)$.

Part I. The maps

$$M \longmapsto \mathrm{Gal}(L/M), \qquad H \longmapsto L^H$$

are mutually inverse, inclusion-reversing bijections between the intermediate fields $\{M : \ K \subseteq M \subseteq L\}$ and the subgroups $\{H : \ H \leq G\}$. Moreover $L : M$ is Galois for every intermediate field M, and

$$[L : M] = |\mathrm{Gal}(L/M)|, \qquad [M : K] = [G : \mathrm{Gal}(L/M)].$$

Part II. For an intermediate field M,

$$M : K \text{ is Galois} \iff \mathrm{Gal}(L/M) \trianglelefteq G,$$

and in this case $\mathrm{Gal}(M/K) \cong \mathrm{Gal}(L/K)/\mathrm{Gal}(L/M)$, the isomorphism being restriction $\gamma \mapsto \gamma|_M$.

Problem 2

Let $\alpha = \sqrt{2+\sqrt{2}}$ (the positive real root). *(In part (b) the statement "K" is a misprint for L.)*

(a) From $\alpha^2 = 2+\sqrt{2}$ we get $(\alpha^2-2)^2 = 2$, so α is a root of

$$f(x) = x^4 - 4x^2 + 2.$$

f is monic and Eisenstein at the prime 2 ($2 \mid -4$, $2 \mid 2$, $2^2 \nmid 2$), hence irreducible over $\mathbb{Q}$. Therefore

$$m_\alpha(x) = x^4 - 4x^2 + 2, \qquad [\mathbb{Q}(\alpha):\mathbb{Q}] = 4.$$

(b) Solving $x^4 - 4x^2 + 2 = 0$ gives $x^2 = 2 \pm \sqrt{2}$, so the roots of m_α are $\pm\alpha, \pm\beta$ with $\beta = \sqrt{2-\sqrt{2}} > 0$ (real, since $\sqrt{2} < 2$). Now

$$\alpha\beta = \sqrt{(2+\sqrt{2})(2-\sqrt{2})} = \sqrt{2}, \qquad \sqrt{2} = \alpha^2 - 2 \in \mathbb{Q}(\alpha),$$

so $\beta = \dfrac{\alpha^2-2}{\alpha} \in \mathbb{Q}(\alpha) = L$. Thus all four roots $\pm\alpha, \pm\beta$ lie in L, and $L = \mathbb{Q}(\alpha) = \mathbb{Q}(\alpha, -\alpha, \beta, -\beta)$ is generated by the roots. Hence L is a splitting field of $m_\alpha(x)$ over $\mathbb{Q}$. Since $\operatorname{char}\mathbb{Q} = 0$, m_α is separable, so $L:\mathbb{Q}$ is Galois and $|\mathrm{Gal}(L/\mathbb{Q})| = [L:\mathbb{Q}] = 4$.

(c) Since m_α is irreducible and β is a root of m_α, there is $\sigma \in \mathrm{Gal}(L/\mathbb{Q})$ with $\sigma(\alpha) = \beta$. We compute σ on all the roots:

$$\sigma(\sqrt{2}) = \sigma(\alpha^2-2) = \beta^2 - 2 = (2-\sqrt{2}) - 2 = -\sqrt{2},$$
$$\sigma(\beta) = \sigma\left(\frac{\sqrt{2}}{\alpha}\right) = \frac{\sigma(\sqrt{2})}{\sigma(\alpha)} = \frac{-\sqrt{2}}{\beta} = -\alpha \qquad (\text{since } \alpha\beta = \sqrt{2}).$$

Hence

$$\sigma: \ \alpha \mapsto \beta \mapsto -\alpha \mapsto -\beta \mapsto \alpha,$$

so σ has order $4 = |\mathrm{Gal}(L/\mathbb{Q})|$. Therefore

$$\mathrm{Gal}(L/\mathbb{Q}) = \langle\sigma\rangle \cong \mathbb{Z}/4\mathbb{Z}.$$

(d) Since $\mathbb{Z}/4\mathbb{Z}$ has exactly three subgroups, $\{e\} \le \langle\sigma^2\rangle \le \langle\sigma\rangle$, there are exactly three intermediate fields by Part I of the Fundamental Theorem. We have $\sigma^2 : \alpha \mapsto -\alpha$, $\beta \mapsto -\beta$, and $\sigma^2(\sqrt{2}) = \sigma(-\sqrt{2}) = \sqrt{2}$. So σ^2 fixes $\sqrt{2}$, and

$$L^{\langle\sigma^2\rangle} \supseteq \mathbb{Q}(\sqrt{2}), \qquad [L^{\langle\sigma^2\rangle}:\mathbb{Q}] = [\mathrm{Gal}(L/\mathbb{Q}):\langle\sigma^2\rangle] = 2 = [\mathbb{Q}(\sqrt{2}):\mathbb{Q}],$$

so $L^{\langle\sigma^2\rangle} = \mathbb{Q}(\sqrt{2})$. The correspondence is:

$H \le \mathrm{Gal}(L/\mathbb{Q})$	L^H	$[L^H:\mathbb{Q}]$
$\{\mathrm{id}\}$	$L = \mathbb{Q}\big(\sqrt{2+\sqrt{2}}\big)$	4
$\langle\sigma^2\rangle = \{\mathrm{id}, \sigma^2\}$	$\mathbb{Q}(\sqrt{2})$	2
$\langle\sigma\rangle = \mathrm{Gal}(L/\mathbb{Q})$	$\mathbb{Q}$	1

So the subfields of L are exactly $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2}) \subset L$. (Since $\mathbb{Z}/4\mathbb{Z}$ is abelian, every subgroup is normal, so every subfield is Galois over $\mathbb{Q}$ by Part II.)

Problem 3

(a) A finite group G is **solvable** if it has a finite filtration

$$\{e\} = G_0 \leq G_1 \leq \cdots \leq G_n = G, \qquad G_i \trianglelefteq G_{i+1},$$

such that every quotient G_{i+1}/G_i is abelian.

(b) Let $f \in K[x]$ be separable with splitting field L over K (char 0). f is **solvable by radicals over** K if $L : K$ is a solvable extension: there is a field $M \supseteq L \supseteq K$ such that $M : K$ is a radical extension, i.e. $M = K(\alpha_1, \ldots, \alpha_k)$ with $\alpha_i^{n_i} \in K(\alpha_1, \ldots, \alpha_{i-1})$ for some integers $n_i \geq 1$. Equivalently, every root of f can be obtained from elements of K using $+, -, \times, \div$ and n-th roots. By Galois' criterion (for a Galois extension $L : K$ in characteristic 0), this holds if and only if $\mathrm{Gal}(L/K)$ is a solvable group.

(c) Let p be a prime and $f(x) = x^5 - p^2x - p \in \mathbb{Q}[x]$, with splitting field L.

Step 1: f is irreducible over $\mathbb{Q}$. f is monic, p divides the coefficients $-p^2$ and $-p$ (and the zero coefficients), and $p^2 \nmid -p$. By Eisenstein's criterion at p, f is irreducible over $\mathbb{Q}$.

Step 2: f has exactly three real roots. We evaluate f at the points $-p, -1, 0, p$ (note $p \geq 2$):

$$\begin{aligned} f(-p) &= -p^5 + p^3 - p < 0 \quad (\text{since } p^5 > p^3), \\ f(-1) &= -1 + p^2 - p = p^2 - p - 1 > 0 \quad (\text{since } p^2 - p - 1 \geq 1 \text{ for } p \geq 2), \\ f(0) &= -p < 0, \\ f(p) &= p^5 - p^3 - p = p(p^4 - p^2 - 1) > 0 \quad (\text{since } p^4 - p^2 - 1 \geq 11 \text{ for } p \geq 2). \end{aligned}$$

By the intermediate value theorem, f has a root in each of the intervals $(-p, -1)$, $(-1, 0)$, $(0, p)$: at least three real roots. On the other hand, $f'(x) = 5x^4 - p^2$ has exactly two real zeros $\pm\sqrt[4]{p^2/5}$, so by Rolle's theorem f has at most three real roots. Hence f has *exactly three* real roots and, since $\deg f = 5$, exactly two non-real roots, which are complex conjugates of each other.

Step 3: $\mathrm{Gal}(L/\mathbb{Q}) \cong S_5$. Identify $G = \mathrm{Gal}(L/\mathbb{Q})$ with a subgroup of S_5 via its action on the 5 distinct roots of f.

- Since f is irreducible of degree 5, G acts transitively on the roots, so the orbit of a root has size 5 and by Orbit–Stabilizer $5 \mid |G|$. By Cauchy's theorem G contains an element ρ of order 5, which is a 5-cycle in S_5.
- Complex conjugation restricts to an automorphism of L (as $L : \mathbb{Q}$ is normal, hence L is stable under conjugation) that fixes the three real roots and swaps the two non-real ones. So G contains a transposition $\tau = (a\, b)$.
- A 5-cycle and a transposition generate S_5 when 5 is prime: replacing ρ by a power ρ^k with $\rho^k(a) = b$ (which is again a 5-cycle) and relabeling, we may assume $\rho = (1\,2\,3\,4\,5)$ and $\tau = (1\,2)$. These generate S_5, since conjugating $(1\,2)$ by powers of ρ yields all $(i\ \ i{+}1)$, and adjacent transpositions generate S_5.

Therefore $G = \mathrm{Gal}(L/\mathbb{Q}) \cong S_5$.

Step 4: S_5 is not solvable. $A_5 \trianglelefteq S_5$ is simple and non-abelian, so the only filtration of A_5 without redundancies is $\{e\} \trianglelefteq A_5$, whose quotient A_5 is not abelian; thus A_5 is not solvable. Subgroups of solvable groups are solvable, hence $S_5 \supseteq A_5$ is not solvable.

Conclusion. The splitting field $L : \mathbb{Q}$ is Galois with non-solvable Galois group S_5. If f were solvable by radicals over $\mathbb{Q}$, then $L : \mathbb{Q}$ would be a solvable extension, and by the Main Theorem ($L : K$ solvable $\Rightarrow \mathrm{Gal}(L/K)$ solvable) $\mathrm{Gal}(L/\mathbb{Q})$ would be solvable, a contradiction. Hence $x^5 - p^2x - p$ is **not solvable by radicals** over $\mathbb{Q}$, for every prime p.